

Relapsing Fever (MONDO:0019633): Comprehensive Disease Characteristics Report

Disease: Relapsing Fever · **MONDO:** MONDO:0019633 · **Category:** Infectious Disease (vector-borne bacterial zoonosis / epidemic infection)

Evidence-source note: Relapsing fever (RF) is an **infectious, non-heritable** disease. Template sections that assume a Mendelian/genetic etiology (causal human genes, pathogenic variants, inheritance pattern, penetrance, karyotype/FISH, carrier screening, genetic counseling) are **Not Applicable to the human host**; the relevant "genetics" is that of the *pathogen* (*Borrelia* antigenic-variation loci) and of *host susceptibility/immune-control genes* studied in mice. Evidence below is drawn from aggregated disease-level resources plus primary human-clinical, model-organism, and in-vitro literature (49 papers reviewed; PMIDs cited throughout).

Summary

Relapsing fever (RF) is an acute, arthropod-borne bacterial infection caused by spirochetes of the genus *Borrelia*, characterized by recurring waves of high fever separated by afebrile intervals. It is an **infectious, non-heritable disease**; the human genome does not "cause" it. The genetics that matter are those of the **pathogen's antigenic-variation loci** and of **murine host-susceptibility genes** used in experimental models. RF exists in two principal epidemiological forms: **louse-borne relapsing fever (LBRF)**, caused by *Borrelia recurrentis* and transmitted by the human body louse (*Pediculus humanus humanus*), and **tick-borne relapsing fever (TBRF)**, caused by numerous species (*B. hermsii*, *B. duttonii*, *B. crocidurae*, *B. hispanica*, *B. persica*, *B. turicatae*, and the emerging hard-tick species *B. miyamotoi*) transmitted by soft ticks (*Ornithodoros* spp.) or *Ixodes* hard ticks.

The **defining mechanistic feature** of the disease — the relapse phenomenon — is driven by **multiphasic antigenic variation of the variable major proteins (Vmps)** on the spirochete outer surface. Through nonreciprocal gene conversion, one of roughly 60 silent *vmp* gene

archives is placed downstream of a single active promoter, allowing the population to serially switch its dominant surface antigen and repeatedly evade the host antibody response. Each bacteremic wave is cleared by a **T-independent, B1b-lymphocyte-derived IgM** response, but escape variants seed the next relapse. This produces the pathognomonic sawtooth fever curve.

Clinically, RF causes high fever, chills, headache, myalgia, hepatosplenomegaly, thrombocytopenia and anemia, and — in severe/untreated cases — ARDS, shock, myocarditis, hemorrhage, and death (up to ~70% untreated in LBRF). Diagnosis rests on **direct visualization of spirochetes on stained (Giemsa/Wright) blood films** during febrile peaks, supplemented by 16S rRNA PCR and GlpQ serology. Treatment is with **tetracyclines (doxycycline), penicillin, or ceftriaxone**, and frequently triggers a **Jarisch-Herxheimer reaction (JHR)** — a TNF- α /IL-6/IL-8-driven cytokine storm coincident with antibiotic-induced spirochete lysis. Prevention relies on vector avoidance, delousing (LBRF), and doxycycline post-exposure prophylaxis (TBRF); no vaccine exists.

Key Findings

Finding 1 — Antigenic variation of variable major proteins (Vmps) drives the relapse phenomenon

The central molecular mechanism of relapsing fever is the multiphasic antigenic variation of surface-exposed **variable major proteins (Vmps)**, which comprise **variable large proteins (Vlp)** and **variable small proteins (Vsp)**. In *B. hermsii*, antigenic variation occurs through a **nonreciprocal gene conversion event** that places one of approximately 60 silent *vmp* genes downstream of a single, transcriptionally active promoter (the "expression site") ([PMID: 29250931](#)). Silent *vmp* gene archives reside on linear plasmids — for example, the 44-kb linear plasmid of *B. duttonii* strain Ly preserves ~21 *vmp* homologues, most rendered non-functional by frameshifts or missing promoters, serving as an archive for serotype switching ([PMID: 12008924](#)). Expression is also controlled by switching between expression sites at the transcriptional level ([PMID: 11083837](#)).

The functional necessity of this system is demonstrated by loss-of-function experiments: **deletion of the expression site or mutation of the flanking cis-acting UHS/DHS inverted repeats renders spirochetes unable to relapse in immunocompetent mice** ([PMID: 29250931](#)), and a **Vmp-null mutant could not relapse and had lower blood densities** ([PMID: 24699793](#)). As the abstract states: "*the spirochetes repeatedly evade the host's acquired immune*

response by undergoing antigenic variation of the variable major proteins (Vmpps) produced on their outer surface. This mechanism prolongs spirochete circulation in blood" (PMID: 24699793). The primary defining quote from the mechanism paper: "In *Borrelia hermsii*, antigenic variation occurs as a result of a nonreciprocal gene conversion event that places one of ~60 silent variable major protein genes downstream of a single, transcriptionally active promoter" (PMID: 29250931).

Finding 2 — Relapsing fever causes high untreated mortality with anemia, ARDS, shock, and JHR as leading complications

A retrospective study of 119 blood-film-confirmed RF patients in a resource-limited Ethiopian setting (mean age 24.1 years, mostly daily laborers and homeless persons) quantified the clinical burden. Leading symptoms were fever (85.5%), chills (67.2%), respiratory distress (54.6%), and altered mental status (46.2%). **Complications occurred in 88.2% of cases** — anemia (74.7%), acute respiratory distress syndrome (69.7%), shock (60.5%), and Jarisch-Herxheimer reaction (36.1%) — with an **in-hospital mortality of 45.4%** (PMID: 41418926). Untreated mortality may reach **70%** (PMID: 41418926).

Clinical feature	Frequency (%)
Fever	85.5
Chills	67.2
Respiratory distress	54.6
Altered mental status	46.2
Any complication	88.2
Anemia	74.7
ARDS	69.7
Shock	60.5
Jarisch-Herxheimer reaction	36.1
In-hospital mortality	45.4

Direct quotes: "Complications were observed in 88.2% of cases, most commonly anemia (74.7%), acute respiratory distress syndrome (69.7%), shock (60.5%), and Jarisch-Herxheimer reaction (36.1%). Mortality was 45.4%" and "Mortality may reach 70% without treatment" (PMID: 41418926). The Jarisch-Herxheimer reaction is "a transient response that occurs within 24 hours of antibiotic treatment for spirochete infections, such as syphilis, leptospirosis, Lyme disease, and relapsing fever, and can present with nausea, fever, chills, rigors, vomiting, hypotension, and skin lesions" (PMID: 40624823).

HPO term suggestions: Fever (HP:0001945), Recurrent fever (HP:0001954), Anemia (HP:0001903), Thrombocytopenia (HP:0001873), Splenomegaly (HP:0001744), Hepatomegaly (HP:0002240), Acute respiratory distress syndrome (HP:0033677), Shock (HP:0031273).

Finding 3 — Anemia and thrombocytopenia arise from spirochete-erythrocyte interaction and macrophage-mediated clearance

In inbred mouse models of *B. hermsii* infection, susceptibility is genetically controlled, and disease progression is associated with **thrombocytopenia and anemia mirroring human infection**. Histological and fluorescence in situ hybridization (FISH) analyses showed that **red blood cells are removed by tissue-resident macrophages in spleen and liver**, and spirochetes were frequently visualized associated with RBCs, supporting a direct spirochete-RBC interaction that leads to bacterial clearance and anemia (PMID: 19995898). The study found: "the progression of the disease in mice was associated with thrombocytopenia and anemia. Histological and fluorescence in situ hybridization (FISH) analysis of infected tissues indicated that red blood cells (RBCs) were removed by tissue-resident macrophages, a process that could lead to anemia. Spirochetes in the spleen and liver were often visualized associated with RBCs." Both innate and adaptive immunity contribute to reducing bacterial burden: "we found that innate immunity contributes significantly to the reduction of bacterial burden" (PMID: 19995898).

Cell types (CL) / anatomy (UBERON): macrophage (CL:0000235), erythrocyte (CL:0000232), spleen (UBERON:0002106), liver (UBERON:0002107).

Finding 4 — *Borrelia miyamotoi* is an emerging hard-tick relapsing fever spirochete

B. miyamotoi represents a paradigm shift: unlike classic soft-tick/louse RF, it is transmitted by *Ixodes ricinus*-complex **hard ticks**, with rodents as reservoirs. It is "an emerging tick-borne pathogen phylogenetically belonging to spirochaetes causing relapsing fever. It is primarily

transmitted by ticks from the Ixodes ricinus complex" (PMID: 34412488). More than 200 human cases have been described: *"To date more than 200 human cases have been described including five cases of meningoencephalitis in immunocompromised patients"* (PMID: 34412488). Presentation is nonspecific (fever, fatigue, chills, headache, myalgia, arthralgia), and infection is treatable with antibiotics. A Minnesota meningoencephalitis case was diagnosed by CSF Gram stain plus sequencing (PMID: 38916722). Diagnosis relies on *"direct (PCR) and indirect diagnosis (glycerophosphoryldiester-phosphodiesterase (GlpQ) serology)"*, and *B. miyamotoi* was found in 2.2% of *I. ricinus* ticks in Alsace, France (PMID: 32303256).

Finding 5 — TBRF in pregnancy causes very high perinatal mortality

A case-control study in rural Tanzania (137 pregnant vs 120 non-pregnant women with TBRF, 1985–1995) documented severe pregnancy outcomes: *"Risk of birth during the attack of TBRF was 58.0%, with an extremely high perinatal mortality of 436 per 1000 births. The total loss of pregnancies including abortions was 475"* per 1000 (PMID: 9351408). Maternal case-fatality was low and similar between groups (1.5% pregnant vs 1.7% non-pregnant), but **pregnant women showed significantly higher spirochete densities**: *"Pregnant women with TBRF show higher densities of spirochetes than non-pregnant women ($p < 0.001$)"* (PMID: 9351408). Delivery during an attack correlated with spirochetemia density ($p < 0.001$) and gestational age ($p < 0.001$); perinatal death was related to low birthweight and low gestational age rather than the degree of spirochetemia. The mechanistic driver of perinatal loss is therefore **preterm delivery precipitated by the febrile attack**.

Finding 6 — LBRF re-emerged in Europe among East African migrants (2015)

Louse-borne relapsing fever (*B. recurrentis*, transmitted by the *Pediculus humanus humanus* body louse — *"Borrelia recurrentis, transmitted by Pediculus humanus humanus, is the etiological agent of louse-borne relapsing fever (LBRF)"* PMID: 26938933) is endemic to the Horn of Africa (Ethiopia, Somalia, Sudan, Eritrea). In 2015, approximately 26–45 imported cases were diagnosed in migrants across Italy, Switzerland, the Netherlands, and Germany — a marked re-emergence after nearly a decade with essentially no imported European cases. A systematic search *"identified 26 cases of LBRF between July and October 2015 in migrants recently arrived in Europe"* (PMID: 27340042). Diagnosis was made by microscopy: *"Multiple spirochetes were visualized on stained blood films which were identified as Borrelia recurrentis by 16S rRNA gene sequencing. All patients recovered after antibiotic treatment with ceftriaxone and/or doxycycline"* (PMID: 27188655). Meningeal involvement with neck stiffness, CSF

pleocytosis and thrombocytopenia was documented and treated with doxycycline plus ceftriaxone (PMID: 26938933). Concurrent malaria and tuberculosis co-infections occurred in half of a Swiss case series (PMID: 27188655).

Finding 7 — TBRF is a zoonosis maintained in *Ornithodoros* soft-tick/small-mammal enzootic cycles

Tick-borne relapsing fever spirochetes "*of genus Borrelia thrive in enzootic cycles involving Ornithodoros spp. (Argasidae) mainly, and rodents*" (PMID: 38321309). The disease is highly region-specific in its species/vector pairings:

Region	<i>Borrelia</i> species	Vector (<i>Ornithodoros</i>)	Reservoir / evidence
West Africa	<i>B. crocidurae</i>	<i>O. sonrai</i> (rodent-burrow)	9.2% (287/3109) small mammals infected where vector present vs 0/1004 where absent; <i>Gerbillus</i> spp. reservoirs (PMID: 26327444)
Iberian Peninsula	<i>B. hispanica</i>	<i>O. erraticus</i> (pigpen)	Blood-meal analysis 46.8% pig, 35.4% human (PMID: 23808979)
Western North America	<i>B. hermsii</i>	<i>O. hermsi</i>	Wild rodents; RF spirochetes in 7.7% Nevada mule deer (PMID: 21995265)
Middle East	<i>B. persica</i>	<i>O. tholozani</i>	Wildlife reservoirs; mainly transstadial transfer (PMID: 33741637)

The West African data are especially clear: "*In areas with O. sonrai, Borrelia infection was demonstrated in 287 of 3109 (9.2%) small mammals tested, and none was documented in 1004 animals tested from other areas*" (PMID: 26327444). The Iberian vector pairing: "*this tick serves as the vector of human tick-borne relapsing fever caused by the spirochete Borrelia hispanica*" (PMID: 23808979).

Finding 8 — The Jarisch-Herxheimer reaction is a TNF- α /IL-6/IL-8 cytokine storm suppressible by anti-TNF- α antibody

In LBRF (*B. recurrentis*), penicillin treatment triggers the JHR with fever, rigors, and hypotension, coincident with 7-, 6-, and 4-fold transient rises in plasma TNF, IL-6, and IL-8, respectively; these cytokine elevations were absent in the 3/17 patients without JHR, and spirochetes cleared from blood ~5 hours after penicillin (PMID: 1569394). The abstract reports *"a seven-, six-, and fourfold elevation of these plasma cytokine concentrations over admission levels was detected, respectively, occurring in transient form coincidental with observed pathophysiological changes of J-HR."*

Causality was established by a randomized double-blind placebo-controlled trial (n=49): sheep **anti-TNF- α Fab given before penicillin reduced JHR with rigors** from 26/29 (controls) to 10/20 (P=0.006), and blunted the rises in temperature (0.8 vs 1.5°C, P<0.001), pulse, blood pressure, and peak IL-6 (17 vs 50 μ g/L) and IL-8 (205 vs 2000 ng/L) (PMID: 8663853). As stated: *"Ten of the 20 patients given anti-TNF-alpha Fab had Jarisch-Herxheimer reactions with rigors, as compared with 26 of the 29 control patients (P = 0.006)."* This provides direct interventional evidence that TNF- α is the proximal mediator of the JHR.

Finding 9 — Contemporary epidemiology: *B. miyamotoi* surveillance in the US and RF burden in West Africa

US surveillance across 9 Northeast/Midwest states (2013–2019) identified **300 *B. miyamotoi* cases** (166 confirmed, 134 possible): *"During 2013-2019, a total of 300 cases were identified through surveillance; 166 (55%) were classified as confirmed and 134 (45%) as possible. Median age of case-patients was 52 years (range 1-86 years); 52% were male. Most cases (70%) occurred during June-September"* (PMID: 37610298). Fever and headache were common, 28% reported recurring fevers, 55% arthralgia, 16% rash, 13% were hospitalized, and there were no deaths. *B. miyamotoi* is present in questing *Ixodes scapularis* across all 67 Pennsylvania counties (positive pools in 38 counties) (PMID: 38686844). In Senegal, metagenomic sequencing of non-malarial febrile illness detected relapsing fever *Borrelia* in **15.5% of cases — the single most common pathogen**: *"Bacteria were the most common, with relapsing fever Borrelia and spotted fever Rickettsia found in 15.5% and 3.8% of cases, respectively"* (PMID: 38272885).

Finding 10 — Resolution of each bacteremic episode requires T-independent IgM produced by B1b lymphocytes

The immune clearance of each relapse wave is a T-independent process. SCID or Rag^{-/-} mice cannot resolve *B. hermsii* infection (they require T/B cells). However, T-cell-deficient TCR^{-/-} and IL-7^{-/-} mice (which lack T cells and follicular B cells but retain B1 and marginal-zone B cells) efficiently cleared the bacterium, and splenectomized mice still cleared it (ruling out a primary marginal-zone B-cell role). *xid* mice (B1-cell deficient) suffered more severe bacteremia, and a selective expansion of the **B1b subset (IgM-high, IgD-/low, Mac1+, CD23-, CD5-)** coincided with resolution: *"a selective expansion of the B1b (i.e., IgM(high), IgD(-/low), Mac1(+) CD23(-), and CD5(-)) cell subset in infected xid mice, which coincided with the eventual resolution of infection"*. Critically, *"mice selectively incapable of secreting IgM, the dominant isotype produced by B1 cells, were completely unable to clear B. hermsii"* (PMID: 12646649).

Cell type (CL): B-1b B cell (CL:0000821); **GO:** immunoglobulin production (GO:0002377), humoral immune response (GO:0006959).

Finding 11 — Dissemination via hijacked plasminogen/plasmin and adhesins; TLR signaling required for protective antibody

RF spirochetes disseminate through host tissues by exploiting the host fibrinolytic system. *B. hermsii* and *B. recurrentis* bind host **plasminogen (Plg)** on their surface and generate **plasmin (Pla)**; spirochetes with Pla activity *"have been shown to degrade extracellular matrix (ECM) components, in addition to digesting fibrin, facilitating bacterial invasion and dissemination"* (PMID: 25914944). *Borrelia* adhesins mediate attachment to cell-surface molecules and to soluble proteins/ECM, cloaking the surface from immune recognition and facilitating tissue colonization (PMID: 21557056). On the host side, controlling infection relies on a rapid humoral response, and beyond B-cell receptor signaling, *"growing evidence suggests that additional signaling by innate immune receptors such as Toll-like receptors is necessary for optimal T cell-dependent and T cell-independent antibody responses"* to *Borreliae* (PMID: 22202086).

GO terms: plasminogen activation (GO:0031639), proteolysis (GO:0006508), Toll-like receptor signaling pathway (GO:0002224).

Finding 12 — Natural rodent reservoirs tolerate RF *Borrelia*; laboratory mice are the principal disease model

North American deer mice *Peromyscus leucopus* and *P. maniculatus* are important natural reservoirs for hard-tick relapsing fever (and several other zoonoses); despite high prevalence of persistent infection they display little pathology, exhibiting both **partial resistance** (limiting pathogen burden) and **infection tolerance** (reducing damage): *"they appear to have partial resistance, which limits the burden of the pathogen. In addition, they display traits of infection tolerance, which reduces the damage of the infection"* (PMID: 27381345). *Peromyscus* are identified as *"important natural reservoirs for several zoonotic diseases of humans: Lyme disease, human granulocytic anaplasmosis, babesiosis, ehrlichiosis, hard tickborne relapsing fever"* (PMID: 27381345). In West Africa, *Gerbillus* spp. serve as reservoirs for *B. crocidurae* (PMID: 26327444). **Inbred laboratory mice** (both immunocompetent and immunodeficient strains) are the standard experimental model, recapitulating relapsing bacteremia, antigenic variation, thrombocytopenia and anemia (PMID: 19995898; PMID: 12646649; PMID: 24699793); guinea pigs are used for tick-transmission competence studies (PMID: 38321309).

NCBI Taxonomy: *Peromyscus leucopus* (txid10041), *Gerbillus* (txid10045), *Mus musculus* (txid10090).

Finding 13 — Prevention rests on vector avoidance/surveillance and doxycycline post-exposure prophylaxis; no vaccine

In TBRF-endemic Israel, prevention among at-risk military personnel is based on *"heightened awareness and risk stratification by active surveillance of tick bites and selective postexposure prophylaxis (PEP) with doxycycline for tick-bitten individuals"* (PMID: 24107216). Across three outbreaks (35 exposed), tick-bite recognition was only ~50% and the attack rate 25–50%; 20% of TBRF cases had no recognized tick bite. After a revised cohort-based policy, *"24 soldiers (including eight with recognized and 16 with unrecognized tick bites) received antimicrobials following the diagnosis of TBRF among their cohorts, and none of these individuals subsequently developed TBRF"* (PMID: 24107216). **No licensed vaccine exists** for relapsing fever; primary prevention is environmental and behavioral (delousing/hygiene for LBRF; tick avoidance and rodent-proofing of dwellings for TBRF).

Section-by-Section Report

1. Disease Information

Overview. Relapsing fever is an acute febrile spirochetal zoonosis/anthroponosis characterized by recurring episodes of fever separated by afebrile intervals, caused by *Borrelia* species. It presents as an undifferentiated febrile illness that is frequently confused with malaria, typhoid, dengue, and leptospirosis.

Key identifiers: - **MONDO:** MONDO:0019633 (relapsing fever) - **MeSH:** Relapsing Fever (D012061) - **ICD-10:** A68 (A68.0 louse-borne; A68.1 tick-borne; A68.9 unspecified) - **ICD-11:** 1C1B - **Not applicable:** OMIM and Orphanet Mendelian-disease identifiers — RF is an infectious, non-heritable disease, so it is not a classic OMIM/Orphanet genetic entry.

Synonyms/alternative names: recurrent fever, famine fever, tick fever, spirochetal fever; louse-borne relapsing fever (LBRF; epidemic relapsing fever); tick-borne relapsing fever (TBRF; endemic relapsing fever); *Borrelia miyamotoi* disease (hard-tick relapsing fever).

Information source. Evidence is drawn from **aggregated disease-level resources** — clinical case series, hospital cohorts, systematic reviews, and experimental animal studies — rather than a single EHR-derived individual-patient dataset.

2. Etiology

Causal factor: infectious. RF is caused entirely by *Borrelia* spirochetes; there is **no genetic (human heritable) etiology**. The primary agents are *B. recurrentis* (LBRF) and multiple TBRF species (*B. hermsii*, *B. duttonii*, *B. crocidurae*, *B. hispanica*, *B. persica*, *B. turicatae*, *B. miyamotoi*, and others).

Environmental / risk factors. Poverty, overcrowding, homelessness, war, famine, and refugee movements favor body-louse infestation and LBRF ([PMID: 27340042](#)). Occupational/recreational exposure to *Ornithodoros*-infested rodent burrows, caves, and rustic cabins drives TBRF. **Pregnancy** is a major risk factor for severe outcomes (higher spirochetemia, high perinatal mortality) ([PMID: 9351408](#)). **Immunocompromise** (e.g., rituximab therapy) predisposes to *B. miyamotoi* meningoencephalitis ([PMID: 34412488](#)).

Genetic risk / protective factors. *Human* — Not established; no validated human susceptibility or protective variants. *Model organism* — susceptibility to *B. hermsii* is genetically controlled across inbred mouse strains via innate-immune genes ([PMID: 19995898](#)). Natural reservoir *Peromyscus* mice carry resistance/tolerance traits that limit pathology ([PMID: 27381345](#)).

Gene–environment interaction. Not applicable in the human-heritable sense. The operative interaction is **pathogen antigenic-variation genetics** × **host adaptive immunity**: host antibody pressure selects for Vmp switch variants, producing the relapse cycle.

3. Phenotypes

The cardinal phenotype is **relapsing fever** — recurring high-fever episodes (present in ~100% of confirmed cases), each lasting ~3 days with afebrile intervals of ~5–7 days, driven by antigenic variation. Additional phenotypes and their characteristics:

Phenotype	Type	Frequency	HPO suggestion
Recurring fever	symptom	~100% (defining)	HP:0001954 (recurrent fever)
Chills/rigors	symptom	~67%	HP:0025143
Headache	symptom	common	HP:0002315
Myalgia/arthralgia	symptom	55% arthralgia (<i>B. miyamotoi</i>)	HP:0003326 / HP:0002829
Thrombocytopenia	lab abnormality	most suggestive lab finding	HP:0001873
Anemia	lab abnormality	74.7% (severe cohort)	HP:0001903
Hepatosplenomegaly	clinical sign	common	HP:0001433
Altered mental status / neurologic	clinical sign	46.2% severe cohort; meningoencephalitis in immunocompromised	HP:0011446
ARDS / respiratory distress	complication	54.6–69.7% (severe cohort)	HP:0033677
Shock	complication	60.5% (severe cohort)	HP:0031273
Petechial rash / hemorrhage	physical manifestation	common in LBRF	HP:0000979

Onset: acute, 2–18 days after exposure. **Severity:** variable — mild self-limiting in *B. miyamotoi* to severe/fatal in untreated LBRF. **Progression:** episodic/relapsing-remitting by definition. **Quality of life:** acute and self-limited if treated; severe morbidity (anemia, ARDS, perinatal loss) and death in untreated/severe cases.

4. Genetic/Molecular Information

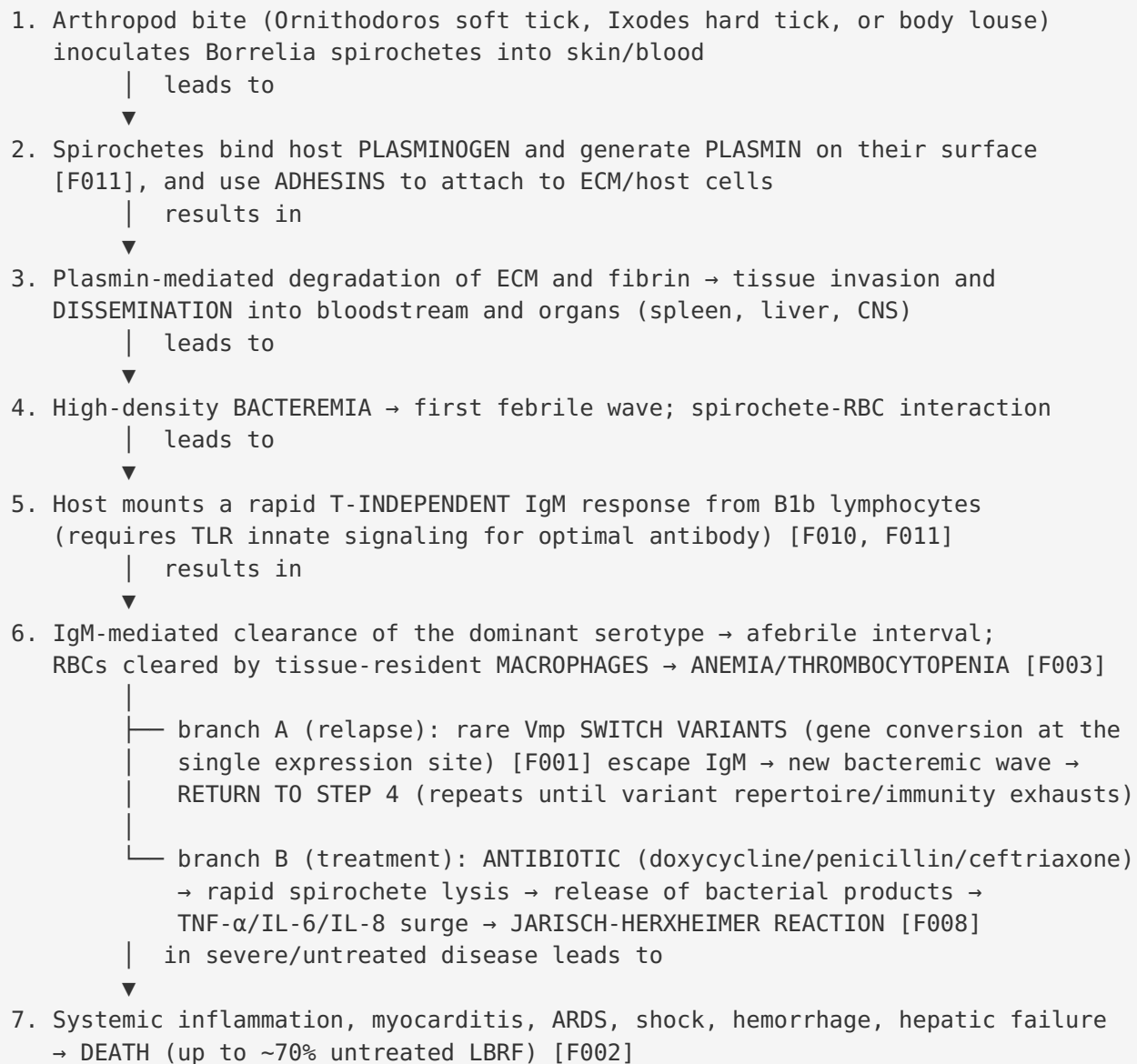
Not applicable to human heritable genetics. No human causal genes, pathogenic variants, modifier genes, or chromosomal abnormalities are associated with relapsing fever.

Pathogen molecular genetics (the operative "genetics"): - Variable major protein (*vmp*) gene families — *vlp* (variable large protein) and *vsp* (variable small protein) — provide ~60 silent archival genes on linear plasmids ([PMID: 12008924](#); [PMID: 11083837](#)). - Single active expression site with flanking *cis*-acting UHS/DHS inverted repeats*; gene conversion into this site produces serotype switching. Deletion/mutation abolishes relapse ([PMID: 29250931](#)). - *glpQ* (glycerophosphoryldiester phosphodiesterase) — a conserved gene absent in *Lyme Borrelia**, used as a serological diagnostic antigen ([PMID: 32303256](#)). - *Vtp*** (variable tick protein) — expressed in the tick vector, invariable at its locus but diverse between strains, and also elicits host antibody.

5. Environmental Information

Infectious agents (taxonomy): *Borrelia recurrentis*, *B. hermsii*, *B. duttonii*, *B. crocidurae*, *B. hispanica*, *B. persica*, *B. turicatae*, *B. miyamotoi* (genus *Borrelia*, family Borreliaceae/Spirochaetaceae). **Vectors:** body louse *Pediculus humanus humanus* (LBRF); soft ticks *Ornithodoros* spp. and hard ticks *Ixodes* spp. (TBRF). **Environmental/lifestyle factors:** poverty, crowding, homelessness, poor hygiene, cold climates promoting layered clothing (lice); exposure to rodent-infested dwellings, caves, and burrows (ticks). No chemical toxin or radiation etiology.

6. Mechanism / Pathophysiology — Ordered Causal Chain



Upstream vs downstream. The **initiating lesion** is vector inoculation (step 1); the **upstream driver of chronicity** is Vmp antigenic variation (step 6, branch A); **downstream effectors** are macrophage-mediated cytopenias, plasmin-driven dissemination, and — during therapy — the TNF-α cytokine cascade of the JHR. Note that branch A is *demonstrated* (loss-of-function mutants cannot relapse); the precise trigger of JHR immunopathology in branch B is *inferred* to be lysis-released bacterial products acting via innate receptors, with TNF-α causality established by the anti-TNF-α RCT.

Molecular pathways / processes: plasminogen activation and proteolysis (GO:0031639, GO:0006508); Toll-like receptor signaling (GO:0002224); humoral immune response / immunoglobulin production (GO:0006959, GO:0002377); cytokine-mediated inflammatory response (TNF- α , IL-6, IL-8; GO:0071356). **Protein dysfunction:** this is a pathogen-driven disease; the key protein events are surface antigen switching (Vmp) and host immune evasion, not host protein misfolding. **Immune involvement:** dominant — antigenic variation for evasion, B1b IgM for control, TNF- α for JHR immunopathology.

Cell types (CL): B-1b B cell (CL:0000821), macrophage (CL:0000235), erythrocyte (CL:0000232), platelet (CL:0000233). **GO biological processes:** antigenic variation / evasion of host immune response, inflammatory response (GO:0006954).

7. Anatomical Structures Affected

- **Primary organs:** blood/bloodstream (site of bacteremia; UBERON:0000178), spleen (UBERON:0002106), liver (UBERON:0002107).
- **Secondary involvement:** central nervous system / meninges (meningoencephalitis, especially *B. miyamotoi* in immunocompromised; UBERON:0001016, UBERON:0002360); lungs (ARDS; UBERON:0002048); heart (myocarditis; UBERON:0000948); placenta and fetus (UBERON:0001987) in pregnancy.
- **Body systems:** hematologic/immune, cardiovascular, respiratory, nervous, hepatobiliary, reproductive.
- **Tissue/cell level:** erythrocytes and platelets (cleared by macrophages), reticuloendothelial macrophages of spleen/liver, B1b lymphocytes.
- **Subcellular:** spirochete surface (outer membrane); no specific host organelle target.
- **Localization:** systemic/bilateral (disseminated bloodborne infection); no lateralization.

8. Temporal Development

- **Onset:** acute, incubation 2–18 days after exposure.
- **Course:** episodic / relapsing-remitting — the defining pattern. In LBRF typically 1 relapse (sometimes more); in TBRF multiple relapses (often 3–10) are characteristic. Each febrile episode lasts ~3 days; afebrile intervals ~5–7 days.
- **Duration:** self-limited over weeks if untreated survivors clear the infection; curative with antibiotics.
- **Critical period:** febrile attacks in pregnancy precipitate preterm delivery — a key window for intervention ([PMID: 9351408](#)). Antibiotic timing determines JHR risk.

9. Inheritance and Population

Inheritance: Not applicable — infectious, not heritable.

Epidemiology / geographic distribution: - **LBRF:** endemic focus in the Horn of Africa (Ethiopia, Somalia, Sudan, Eritrea); epidemic potential in war/famine/refugee settings; 26–45 imported European cases in 2015 ([PMID: 27340042](#); [PMID: 27188655](#)). - **TBRF:** globally distributed enzootic foci (West Africa, Iberia, Middle East, western North America) with region-specific species/vectors ([PMID: 38321309](#); [PMID: 26327444](#)). - **Burden:** relapsing fever *Borrelia* was the most common pathogen (15.5%) in non-malarial febrile illness in Senegal ([PMID: 38272885](#)). - **B. miyamotoi (US):** 300 surveillance cases (2013–2019), median age 52, 52% male, 70% June–September ([PMID: 37610298](#)).

Demographics: all ages affected; severe outcomes in pregnant women, the malnourished, and immunocompromised. Sex ratio roughly balanced (52% male in US *B. miyamotoi* series).

10. Diagnostics

- **Gold standard:** direct microscopy — spirochetes visualized on **Giemsa/Wright-stained thick and thin blood films** drawn during febrile peaks ([PMID: 27188655](#)); thrombocytopenia is the most suggestive laboratory finding.
- **Molecular: 16S rRNA gene PCR/sequencing** for species identification ([PMID: 27188655](#)); qPCR for *B. miyamotoi* ([PMID: 38686844](#)); metagenomic sequencing ([PMID: 38272885](#)).
- **Serology: GlpQ (glycerophosphoryldiester-phosphodiesterase) serology** distinguishes RF from Lyme *Borrelia* ([PMID: 32303256](#)).
- **CSF:** Gram stain + sequencing for meningoencephalitis ([PMID: 38916722](#)); pleocytosis in meningeal involvement ([PMID: 26938933](#)).
- **Differential diagnosis:** malaria (co-infection common), typhoid, dengue, leptospirosis, spotted-fever rickettsiosis.
- **Genetic testing / newborn screening / omics diagnostics:** Not applicable (infectious disease).

11. Outcome/Prognosis

- **Mortality:** untreated LBRF up to ~70%; severe hospitalized cohort 45.4% in-hospital mortality ([PMID: 41418926](#)); with antibiotic treatment, case-fatality falls to <5%. No deaths in the milder US *B. miyamotoi* series ([PMID: 37610298](#)).

- **Complications:** anemia, ARDS, shock, JHR, myocarditis, hemorrhage, hepatic failure, meningoencephalitis, and — in pregnancy — abortion/preterm birth with perinatal mortality of 436/1000 (PMID: 9351408).
- **Prognostic factors:** promptness of antibiotic therapy, pregnancy, high spirochetemia density, immunocompromise, malnutrition, and co-infection.
- **Recovery:** complete recovery is expected with timely treatment; the disease is self-limited/curable rather than chronic.

12. Treatment

- **First-line pharmacotherapy: tetracyclines — doxycycline** (NCIT:C299); **penicillin** (NCIT:C716); **ceftriaxone** (NCIT:C1638) for severe/CNS disease. Erythromycin/macrolides are alternatives (e.g., in children, pregnancy). Patients recovered after ceftriaxone and/or doxycycline (PMID: 27188655); doxycycline + ceftriaxone for meningeal involvement (PMID: 26938933).
- **Jarisch-Herxheimer management:** anticipate JHR within 24 h of first dose; supportive care (fluids, antipyretics, hemodynamic monitoring). **Pre-treatment anti-TNF- α antibody reduces JHR severity** in trials (PMID: 8663853) — a mechanistic proof of concept, not yet standard care.
- **Supportive care:** transfusion for anemia, oxygen/ventilation for ARDS, vasopressors for shock.
- **Advanced/targeted/gene/cell/RNA therapies; pharmacogenomics:** Not applicable.

13. Prevention

- **Primary prevention:** LBRF — **delousing**, sterilizing/laundering clothing, pediculicides, improved personal hygiene; TBRF — avoidance of rodent-infested/tick-infested dwellings, rodent-proofing, acaricides, permethrin-treated clothing.
- **Secondary prevention:** active surveillance of tick bites and early diagnosis in endemic settings.
- **Post-exposure prophylaxis: doxycycline PEP** effective for TBRF; cohort-based PEP prevented all subsequent cases in an Israeli military study (PMID: 24107216).
- **Immunization: no vaccine exists** (antigenic variation is a fundamental barrier to vaccine development).
- **Public health:** outbreak control through delousing campaigns, sanitation, health education, and vector control.

14. Other Species / Natural Disease

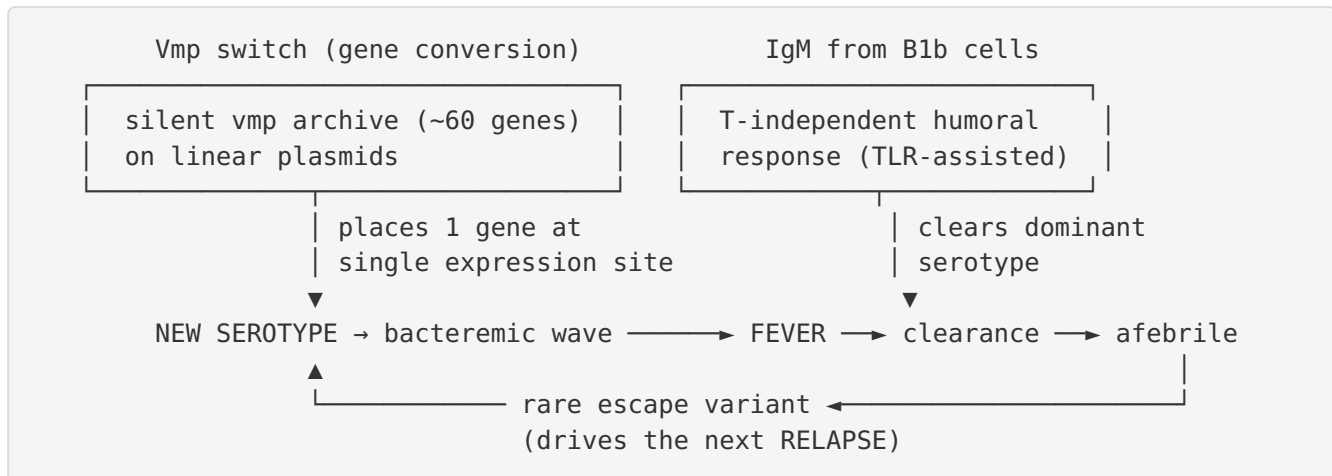
- **Taxonomy of hosts:** *Peromyscus leucopus* / *P. maniculatus* (NCBI txid10041/10042), *Gerbillus* spp. (txid10045), *Mus musculus* (txid10090), *Odocoileus hemionus* (mule deer). Reservoir competence with minimal pathology in *Peromyscus* (PMID: 27381345).
- **Zoonotic potential:** TBRF is a **true zoonosis** maintained in *Ornithodoros*/rodent enzootic cycles (PMID: 38321309); LBRF (*B. recurrentis*) is essentially an **anthroponosis** (human–louse–human cycle).
- **Comparative biology:** RF spirochetes infect wild rodents, deer (7.7% Nevada mule deer; PMID: 21995265), and — for related species such as *B. anserina* and *B. theileri* — birds and cattle. Reservoirs tolerate infection via resistance + tolerance mechanisms (PMID: 27381345).

15. Model Organisms

- **Principal model: inbred laboratory mice** (*Mus musculus*), both immunocompetent and immunodeficient (SCID, Rag^{−/−}, TCR^{−/−}, IL-7^{−/−}, *xid*), which recapitulate relapsing bacteremia, antigenic variation, thrombocytopenia, and anemia and have been used to dissect the B1b/IgM clearance mechanism and genetic susceptibility (PMID: 12646649; PMID: 19995898).
- **Genetic models:** *Borrelia* mutants — Vmp-null, expression-site deletions, UHS/DHS repeat mutants — used to prove antigenic-variation necessity (PMID: 29250931; PMID: 24699793).
- **Other models:** guinea pigs for tick-transmission competence (PMID: 38321309); *Peromyscus* reservoir studies for resistance/tolerance biology (PMID: 27381345).
- **Recapitulation/limitations:** mouse models faithfully reproduce the relapse cycle and immune control but do not fully capture human severe complications (ARDS, JHR severity, pregnancy loss).

Mechanistic Model / Interpretation

The unifying concept of relapsing fever is an **evolutionary arms race between pathogen antigenic variation and host antibody immunity**, played out in cycles measured in days:



Every febrile peak corresponds to a high-density bacteremic wave of a single dominant Vmp serotype; every trough corresponds to IgM-mediated clearance of that serotype. Because gene conversion continually generates rare escape variants, the process repeats until the variant repertoire and cumulative immunity exhaust it. Superimposed on this cycle are (i) **plasmin/adhesin-mediated dissemination** enabling organ invasion, (ii) **macrophage-mediated erythrocyte/platelet clearance** producing the characteristic cytopenias, and (iii) the **TNF- α /IL-6/IL-8 Jarisch-Herxheimer cascade** unleashed when antibiotics lyse the spirochetes. This model explains the clinical hallmarks (sawtooth fever, anemia, thrombocytopenia, treatment-associated crises) as direct consequences of a single organizing principle — antigenic variation under antibody selection.

Evidence Base

Finding	Key PMIDs	Contribution
Antigenic variation mechanism	29250931 , 24699793 , 12008924 , 11083837	Define gene conversion, expression site, cis-elements; loss-of-function proves relapse dependence
Clinical burden/mortality	41418926 , 40624823	Quantify complications, 45.4% mortality, define JHR
Anemia/macrophage clearance	19995898	FISH evidence of macrophage RBC clearance; innate + adaptive control
<i>B. miyamotoi</i> emergence	34412488 , 38916722 , 32303256	Hard-tick RF, CNS disease, PCR/GlpQ diagnosis
Pregnancy outcomes	9351408	Perinatal mortality 436/1000; higher spirochetemia
LBRF re-emergence	27340042 , 27188655 , 26938933	2015 European migrant cases, diagnosis, treatment
Enzootic cycle	38321309 , 26327444 , 23808979 , 21995265 , 33741637	Region-specific <i>Ornithodoros</i> /rodent cycles
JHR cytokine mechanism	8663853 , 1569394	RCT anti-TNF- α suppresses JHR; TNF/IL-6/IL-8 surge
Epidemiology	37610298 , 38686844 , 38272885	US surveillance; West African burden
B1b/IgM clearance	12646649	IgM essential; B1b expansion resolves bacteremia
Dissemination/TLR	25914944 , 21557056 , 22202086	Plasmin/adhesins; TLR required for antibody
Reservoirs/models	27381345	<i>Peromyscus</i> resistance + tolerance
Prevention/PEP	24107216	Doxycycline PEP; no vaccine

Evidence source types: human clinical (case series, cohorts, RCT of anti-TNF- α , surveillance); model organism (mouse genetics, *Borrelia* mutants, *Peromyscus*, guinea pigs); molecular/in vitro (plasmin binding, gene-conversion genetics).

Limitations and Knowledge Gaps

1. **No primary dataset was analyzed** — this report synthesizes published literature; no de novo statistical testing of raw data was performed. Effect sizes are those reported by source studies.
2. **Severe-cohort mortality (45.4%) is not generalizable** — it derives from a single resource-limited hospital cohort of the sickest, blood-film-positive patients and overestimates population case-fatality where treatment is available.
3. **Species heterogeneity** — RF spans many *Borrelia* species with differing virulence (e.g., mild *B. miyamotoi* vs lethal *B. recurrentis*); pooling clinical features across species obscures species-specific behavior.
4. **Human host genetics unexplored** — no GWAS or candidate-gene studies define human susceptibility/protective loci; all host genetic evidence is murine.
5. **Anti-TNF- α for JHR** remains a proof-of-concept from one RCT and is not standard practice; optimal clinical JHR mitigation is unresolved.
6. **Global burden is underestimated** — RF is frequently misdiagnosed as malaria; true incidence/prevalence figures are lacking for most endemic regions.
7. **Vaccine development** is fundamentally hampered by antigenic variation; no candidate has advanced.

Proposed Follow-up Experiments / Actions

1. **Species-stratified clinical meta-analysis** — pool case series by *Borrelia* species to derive species-specific complication rates, JHR risk, and case-fatality with confidence intervals.
2. **Human immunogenetic study** — investigate whether human variants in innate-immunity genes (TLRs, complement) or hemoglobinopathies modulate RF severity, mirroring the murine susceptibility findings ([PMID: 19995898](https://pubmed.ncbi.nlm.nih.gov/19995898/)).

3. **Prospective anti-cytokine JHR trial** — evaluate targeted TNF- α (or IL-6) blockade at antibiotic initiation in LBRF, building on [PMID: 8663853](#), with hemodynamic and cytokine endpoints.
 4. **Conserved-antigen vaccine screen** — target invariant surface molecules (e.g., GlpQ, Vtp-conserved regions, factor-H-binding proteins) to circumvent Vmp antigenic variation.
 5. **Enhanced surveillance & rapid diagnostics** — deploy multiplex PCR/metagenomic panels and GlpQ point-of-care serology in malaria-endemic Africa to correctly attribute non-malarial febrile illness ([PMID: 38272885](#)).
 6. **Pregnancy-focused management protocol** — study fetal monitoring and prophylactic strategies to reduce attack-triggered preterm delivery ([PMID: 9351408](#)).
 7. **One Health reservoir mapping** — characterize *Ornithodoros*/rodent enzootic foci and *B. miyamotoi* hard-tick range expansion to predict emergence risk.
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Conclusion

Relapsing fever is an infectious, non-heritable, arthropod-borne spirochetosis whose defining relapse phenomenon is produced by multiphasic **Vmp antigenic variation via gene conversion**, which repeatedly evades a **T-independent, B1b-cell IgM** response that clears each bacteremic wave. It is diagnosed by blood-smear microscopy (with PCR and GlpQ serology) and cured with doxycycline, penicillin, or ceftriaxone — treatment that frequently provokes a **TNF- α -driven Jarisch-Herxheimer reaction**. Untreated louse-borne disease can be highly lethal (up to ~70%). All human-heritable genetics sections are Not Applicable; the operative genetics are those of the pathogen's antigenic-variation loci and of murine host-susceptibility genes.
